

General introduction to the application of the Daisy model

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June 30, 2026.

Abstract

This introduction to the application of the Daisy model comprises minor 12 exercises. Exercise 1 illustrates how to set up a simple simulation using minimum information on the simulated system and how to extract simulation results. Exercises 2 & 3 are dealing with soil hydraulic properties and discretization of the soil profile. Exercise 5 is introducing irrigation. Exercise 4 is dealing with tile drains and the calibration of the interface to the groundwater. Exercise 6 & 7 are introducing the simulation of rotations and the use of organic fertilizers. Exercise 8 is illustrating the importance of introducing a “warm-up” period. In Exercise 9 the MIT-model (Mineralization-Immobilization Turnover) is calibrated. Exercise 10 builds a Daisy weather file. In Exercise 11 an organic fertilizer is parameterised. In Exercise 12 a crop is parameterised.

A technical description of the Daisy model is given by [Abrahamsen and Hansen, 2000] and [Hansen, 2002].

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Introduction

In order to run a Daisy simulation the model needs parameterisation. The required information is parsed on to the model through setup-files (or `.dai` files), which are written in a special input-language. Daisy stores information in an internal database. The information parsed on to the model need not occur in any special order. The Daisy code contains a partial parameterisation, i.e. a number of parameters is given default values at initialisation. However, it is always possible to overwrite the default parameterisation. Information on soil horizons is stored in the internal database by the `horizon`-component:

```
(defhorizon "Ap JB6" FA03
  "An optional description."
  ;; More parameters
  (clay 15.5 [%]))
```

where `defhorizon` is short for define horizon. Other similar expressions are `defcolumn` (defining a soil column including the soil surface and the bioclimate above it), `defcrop` (defining crop growth parameters), `defam` (defining added matter, e.g. fertilizers), `defaction` (defining an action list of management actions), and `deflog` (defining output-variables). "*Ap JB6*" is the name which the parameterisation will have in the internal database and which later can be referred to when one wants to retrieve the stored information. The `FA03` keyword retrieves the structure of the internal database and, hence, it specifies the type of information that now can be expected and it also includes default values given to any of the parameters of the horizon. Alternatively, the name of an existing element in the database (a previously defined horizon) could be stated instead of the expression `FA03`¹. The expression "`(clay 15.5 [%])...`" refers to an attribute list (an attribute list is an unordered set of (*name*, *value*) pairs), which contains the parameters to be added or changed as compared to the selected database element (default or a previously defined horizon). In the present example the component "horizon" has a member "clay" of type "number" and furthermore the value 15.5 is assigned to this member. The square brackets after the value 15.5 is used to give the dimension of the number. In the present example the clay content is given as percent. If the brackets are left out, Daisy will use a default dimension, which may not be what you think. So don't leave out the dimension.

It is noted that parentheses are used to group information e.g. `(clay 0.155 [%])`. Similarly all the information defined by the `defhorizon` statement and stored under the name "*Ap JB6*" are embedded in parentheses. Parentheses are the main structural element of the Daisy language syntax.

Furthermore it is noted that the name "*Ap JB6*" is imbedded in quotation marks. Names may always be imbedded in quotation marks, but it is only necessary if the name contains special symbols including white spaces. The only special symbol, which can be used without the use of quotation marks, is the underscore.

¹`(defhorizon "Ap Askov" "Ap JB6" (humus 1.5 [%]))` this statement overwrites the humus content but retains all other information stored in "*Ap JP6*".

Comments, which are ignored by the parser, can be included in Daisy-setup-files by use of semicolons (;). The parser ignores any text occurring after a semicolon until end-of-line. The only exception is when the semicolon is imbedded in quotation marks, i.e. is a part of a name.

A comprehensive description of the Daisy input-language is given in *Daisy Program Reference Manual* [Abrahamsen, 2011a].

1 Writing a Daisy Setup File

The objective of this exercise is to illustrate how to write a Daisy-setup-file, run a simple simulation, and extract simulation results.

When writing a Daisy-setup-file it is practical first to define computational environment. The working folder is by default the folder where the Daisy-setup-file is stored. However, output files can be routed to a sub-folder by use of the command (`log_prefix "E1/"`) (see below). At installation Daisy stores external library-files in prededined folders, where Daisy by default will search for the information. Box 1.1 shows how external library-files including the weather file are retrived by the model. When the Daisy code is running it writes a log-file (`daisy.log`), where all messages from the simulation, including error messages, is stored in the working folder.

Box 1.1: Daisy-setup-file header

```
;; Including external library-files
(input file "tillage.dai")
(input file "crop.dai")
(input file "fertilizer.dai")
(input file "log.dai")

;; Weather data
(weather default "dk-taastrup.dwf")
```

The external library-files are written in the same Daisy input-language as the Daisy-setup-file. In fact any information, which is supplied by the Daisy-setup-file, could be moved to an external file, which then could be included by a input file statement. On the other hand any information supplied by the external library-files could also be written directly to the Daisy-setup-file. The use of external library-files is just a convenient way of organising ones data. There is only one hard rule that must be obeyed and that is that any information that is used must be defined before any reference is made to it.

Weather data are driving variables in the simulation and is supplied to the model through a special file written in a special format (daisy weather format or `dwf`). This format shall be explained later.

Basic soil horizon information is given in table 1.1. This information is transferred to the internal library of the model using `defhorizon` as shown in box 1.2.

Using `defhorizon` many more parameters can be defined, see *Daisy Program Reference Manual* [Abrahamsen, 2011a]. When these parameters are not supplied by the `defhorizon`-statements then the model uses either predefined default values or pedotransfer functions to estimate the missing parameters. E.g., turnover rates of organic matter have been assigned default values, soil thermal properties are estimated by a modified de Vries model [Hansen, 2002], and soil hydraulic properties are estimated by the HYPRES pedotransfer function [Wösten et al., 1999].

Table 1.1: Basic soil horizon parameters.

Depth cm	Dry bulk density g/cm ³	Humus	Clay <2 μ m	Silt	Sand >50 μ m
00 – 30	1.53	0.026	0.113	0.277	0.584
30 – 80	1.51	0.005	0.235	0.253	0.507
80 –	1.57	0.002	0.244	0.283	0.471

Box 1.2: Defining a horizon.

```
;; Defining a soil horizon
(defhorizon "Ap F.S.L." FA03
  "Data from Jacobsen, 1989"
  (dry_bulk_density 1.53 [g/cm^3])
  (clay 11.3 [%])
  (silt 27.7 [%])
  (sand 58.4 [%])
  (humus 2.6 [%])
  (C_per_N 11.0 [g C/g N]))
```

Based on horizon information a column can now be build. In principle the column comprises the bioclimate (upper boundary, exchange with the atmosphere), the soil profile itself, and a groundwater condition (lower boundary, exchange with the groundwater). However the bioclimate component is fully parameterised by defaults, and hence it is not necessary to include it in the setup. A column parameterisation based on minimum requirements is shown in box 1.3. Firstly, it is noticed, that

Box 1.3: Defining a column.

```
;; Parameterisation of column (Fine sandy loam)
(defcolumn "Fine sandy loam" default
  (Soil (MaxRootingDepth 100 [cm])
    (horizons ( -30 [cm] "Ap F.S.L.")
      ( -80 [cm] "B F.S.L.")
      ( -400 [cm] "C F.S.L.")))
  (OrganicMatter original (init (input 3000 [kg C/ha/y])))
  (Groundwater deep))
```

the column is given the name "Fine sandy loam", secondly, that the parameterisation comprises two sections, a soil section and a groundwater section, as the bioclimate section is omitted. The soil section contains information on the maximum rooting depth in the soil and the soil horizons of the profile. The A-horizon extends down to 30 cm depth, the B-horizon is found in the interval 30-80 cm and C-horizon starts at 80 cm depth and continues down to at least 400 cm depth, where the calculations stop.

The organic matter section gives information used in initialization of the MIT-model (*Mineralization-Immobilization-Turnover* model). The figure 3000 kg C/ha/y indicated the average annual input of organic carbon in the soil in the decade before the start of the simulation. The carbon input comprises

carbon in organic fertilizer, plant residuals, and rhizodeposition. The model contains a procedure that automatically parameterize the MIT-model based on this information.

The groundwater section tells that the position of the groundwater table is located deep below the 400 cm limit of the calculations. This means that a free drainage can be selected for the soil water dynamics calculations. Again, when only this minimum information is supplied then the model makes extensive use of default parameterisation.

The next step is to define a list of actions performed in order to grow spring barley; in this case the list is named `SBarley_management`, box 1.4. It is noted that the action list comprises of activities

Box 1.4: Spring barley management.

```
;; Defining an action list related to management
(defaction SBarley_management activity
  (description "Management plan for spring barley.")
  (wait_mm_dd 3 20)
  (plowing)
  (wait_mm_dd 4 15)
  (seed_bed_preparation)
  (sow "Spring Barley")
  (wait_mm_dd 4 20)
  (fertilize (N25S (weight 95 [kg N/ha])))
  (wait (or (crop_ds_after "Spring Barley" 2.0) ;Ripe
            (mm_dd 9 1)))
  (harvest "Spring Barley"
    (stub 8 [cm])
    (stem 0.70 [])
    (leaf 0.70 [])))
```

viz. to wait, to plow, to make a seedbed preparation, to sow, to fertilize and to harvest. The two tillage operations, `plowing` and `seed_bed_preparation`, are in fact lists of primitive actions found in the library “`tillage.dai`”. The command `sow` takes one parameter, which is the name of crop that is to be sown; in this case the name is “`Spring Barley`”. “`Spring Barley`” is the name of a list of crop parameters, which is found in the library “`crop.dai`”, or in fact the parameter list is found in a file which is included (input file) in the `crop.dai`-file. The command `fertilize` takes up to arguments. The first argument comprises, a name, which in the present case is `N25S`, and an attribute list. However, only the weight parameter, (`weight 95 [kg N/ha]`), is given here, the remaining of the required parameters are to be found in the library file “`fertilizer.dai`”. The first argument of command `harvest` is the name of the crop to be harvested, and then follows a list of attributes specifying the stubble length (here 8 cm) and the fraction of straw (`stem 0.70 []`) and leaf (`leaf 0.70 []`) removed from the field at harvest. Defaults are in this case that the stubble length is 0 cm and that all straw and leaf are removed.

Wait statements are used in order to govern the timing of the other management actions. Two types of wait statements are used in the present action list. The first statement tells the model to wait until a certain day in the year defined by month number and day number in the month, e.g. (`wait_mm_dd 3 20`). The second type waits until a certain condition becomes true. The `or`-statement becomes true when one of its conditions becomes true. The `crop_ds_after` statement takes two arguments; the first is the name of the considered crop and the second is a boundary development state. If the development

stage of the crop is greater or equal to the boundary development stage then the statement is true. The crop development stage is 0 at emergence, 1 at flowering, and 2 at maturing. Hence the statement gets true when the spring barley is ripe. The second condition of the or-statement becomes true when the date is 1st of September (`mm_dd 9 1`).

Now everything is ready so we can actually begin a simulation. As shown in box 1.5 we activate a column, select a starting date and an ending date, and we select an already defined agricultural management practice.

Box 1.5: Activate simulation.

```
;; Selecting a previously defined column.
(column 'Fine sandy loam')

;; Start and end of simulation.
(time 1993 1 1)
(stop 1994 1 1)

;; Selecting the agricultural management and the end of simulation
(manager activity
  SBarley_management
```

However, we have not selected any output. This is done with the output statement as shown in box 1.6. In the present case we only make use of predefined output logs that can be found in the external library file “`log.dai`”.

First we write a description of the simulation that will appear in all output files. Then we tell the model to store the output files in a sub-folder to the actual working folder. This is done by the command (`log_prefix "E1/"`), which will place all output files in the subfolder (`E1`). You must create the sub-folder before starting the simulation. When `harvest` is selected then the harvested yields are written to the output-file “`harvest.dlf`”. The extension `.dlf` indicates that the file is written in Daisy-log-format, which is a format that easily can be opened in a spreadsheet. Furthermore it indicates that the output can be inspected by use of the Daisy output-viewer (`ShowDaisyOutput.exe`). The only exception from the latter is the `harvest.dlf`-file; all other output files can be viewed by the output-viewer. Furthermore “`Crop Production`” is selected. In this case daily (default) values of variables pertaining to crop production is written to “`crop_prod.dlf`”. It is noted that the water and nitrogen balances is of the upper 100 cm (the command: (`to -100 [cm]`)) and that both the output frequency daily and monthly is selected (the `when`-command) and that the different output frequencies are written to different output-files (the `where`-command). In addition some profile data are selected as daily-values.

Fill in the necessary information in the Daisy-setup-file `Exercise01.dai` and run a simulation either by activating the model within TextPad (go to Tools and select Daisy) or by dragging and dropping the setup-file at the Daisy executable. Inspect the created output-files by use of the output-viewer and an EXCEL-spreadsheet. Construct a water balance, a mineral nitrogen balance, an ammonium balance and a nitrate balance.

Box 1.6: Select output files.

```
;; Description that will occur in all output files
(description "Spring Barley; Soil: Fine sandy loam; Weather: Taastrup")
(log_prefix "E1/")
(output harvest
  "Crop Production"
;;   Water balance of the upper 100 cm
  ("Soil water" (to -100 [cm])(when daily)
    (where "Daily_WB.dlf"))
  ("Soil water" (to -100 [cm])(when monthly)
    (where "Monthly_WB.dlf"))
;;   Nitrogen balance of the upper 100 cm
  ("Soil nitrogen" (to -100 [cm])(when daily)
    (where "Daily_NB.dlf"))
  ("Soil nitrogen" (to -100 [cm])(when monthly)
    (where "Monthly_NB.dlf"))
;;   Soil profile data
  ("Soil Water Content" (when daily))
  ("Soil Water Potential"(when daily))
  ("Soil NO3 Concentration"(when daily)))
```

2 Setting Hydraulic Properties

The objective of this exercise is to utilize measured soil hydraulic properties, e.g. the soil water retention curve and the hydraulic conductivity curve. Based on measurements soil water retention and unsaturated hydraulic conductivity table 2.1 has been prepared.

Table 2.1: Mualem-van Genuchten parameters of soil hydraulic properties based on data from [Jacobsen, 1989]. Parameter fitting is done by using the RETC software [Genuchten et al., 1998]

Depth cm	θ_s	θ_r	α cm^{-1}	n	K_s cm/h
00–30	0.403	0.000	0.0385	1.211	7.52
30–80	0.421	0.000	0.2605	1.135	14.5
80–	0.401	0.000	0.0570	1.131	1.65

The soil hydraulic properties are a part of the soil horizon. The Daisy code is able to use a number of different descriptions or models of soil hydraulic properties [Abrahamsen, 2011a]. The soil hydraulic properties parameters are written to the Daisy-setup-file in the `defhorizon` section as illustrated in box 2.1.

`M_vG` is the name of the selected hydraulic model (Mualem-van Genuchten model) and the following attribute list is the parameterisation of this model.

Compare the water and nitrogen balance results obtained with the parameterised Mualem–van Genuchhten model with the corresponding results obtained with the pedotransfer function.

Box 2.1: Hydraulic parameters

```
(defhorizon "Ap F.S.L." FA03
  (description "Data from O.H. Jacobsen (1989)")
  ;; Insert other parameters here.
  (hydraulic M_vG
    (Theta_res 0.0 [%])
    (Theta_sat 40.3 [%])
    (alpha 0.0385 [cm-1])
    (n 1.211 [])
    (K_sat 7.52 [cm/h])))
```

3 Calculating Hydraulic Properties

Measured basic soil data for a coarse sandy soil is shown in table 3.1 through table 3.3. Use the RETC software to estimate the necessary parameters in order to build a parameterisation of this soil.

Table 3.1: Basic soil horizon parameters of a coarse sandy soil [Jacobsen, 1989].

Depth cm	Dry bulk density g/cm ³	Humus	Clay <2 μ m	Silt	Sand >50 μ m
00–30	1.49	0.023	0.039	0.072	0.866
30–	1.49	0.004	0.029	0.023	0.944

Table 3.2: Soil water retention data [Jacobsen, 1989].

Depth cm	Porosity (h=0cm)	pF 1.0	pF 1.2	pF 1.7	pF 2.0	pF 2.2	pF 2.7	pF 4.2
00–30	0.427	0.375	0.364	0.211	0.168	0.149	0.119	0.050
30–	0.436	0.369	0.343	0.097	0.073	0.059	0.045	0.020

Table 3.3: Soil hydraulic conductivity data [cm/h] [Jacobsen, 1989].

Depth cm	Saturation (h=0cm)	pF 1.0	pF 1.2	pF 1.4	pF 1.7
00–30	21.7	10.03	7.18	3.95	0.032
30–	39.5	4.59	2.93	0.15	0.002

So far we have used the default discretization in our simulations. Introduce your own discretization and run a simulation. It is hard rule that numeric layers may not cross horizon boundaries. It is a rule of thumb that the size of a numeric layer should not exceed twice the dispersivity of the soil (the dispersivity is a member of the soil as defined within the column and its default value is 5 cm). The numeric layers are defined by their lower boundary (zplus) starting from the top. An example of a discretization is shown in box 3.1. In box 3.1 the specification of the dispersivity might as well be omitted as it only specifies the default value.

Does the crop in this example experience water stress?

Box 3.1: Discretization

```
(defcolumn "Coarse sand" default
;; Insert other parameters here.
(Soil (MaxRootingDepth 50 [cm])
      (dispercivity 5 [cm])
      (horizons ( -30 [cm] "Ap C.S")
                 ( -320 [cm] "C C.S"))))
(Movement vertical
  (Geometry (zplus -2 -4 -7 -10 -13 -19 -22 -25 -27 -30 -35 -40 -45 -50
                 -55 -60 -65 -70 -75 -80 -85 -90 -95 -100 -120 -130
                 -140 -150 -160 -170 -180 --190 -200 -210 --220 -230
                 -240 -250 -260 -270 -280 -290 -300 -310 -320 [cm]))))
```

4 Groundwater

The objective of this exercise is to introduce tile drains and the calibration of the interface to the groundwater.

How tile drains can be introduced in the simulation is described in *Daisy Tutorial*, section 4.4. The soil and the management are characterised as in exercise 2. We know the distance between the drainpipes are 18 m (default value) and that the pipes are located at 110 cm depth (default value). Assume further that an aquitard of thickness 2 m (default value) is located at 2.20 – 4.20 m depth and that the aquitard is overlaying an aquifer, which is characterised by a pressure potential of 200 cm at the top of the aquifer (the interface between the aquitard and the aquifer). Measurement shows that approximately 2/3 of the net precipitation in 1993 was lost through the drainpipes. Calibrate the hydraulic conductivity of the aquitard so that the simulated value matches the measured value.

In the present case the lowest soil horizon ends at 2.2 m depth (i.e. the top of the aquitard). The new lower boundary condition is introduced by entering

```
(Drain lateral)
(Groundwater aquitard (K_aquitard 0.01 [cm/h]))
```

instead of

```
(Groundwater deep)
```

The numerical value assigned to `K_aquitard` is the value used for the calibration.

5 Irrigation

The objective of this exercise is to introduce irrigation in the management practice.

Develop an irrigation strategy for the spring barley grown on a coarse sandy soil (exercise 3). The *Daisy Tutorial* section 5.5 describes how this can be done, and further information can be obtained from the *Daisy Program Reference Manual*.

6 Rotations

The objective of this exercise is to introduce the simulation of a crop rotation.

Assume the same setup for the column as in exercise 4.

A common crop rotation that is used at arable farms as well pig farms are: spring barley, winter barley, winter rape, winter wheat and winter wheat. A suggestion for management practices typical for Danish conditions pertaining to this rotation is shown in table 6.1.

Table 6.1: Management practices of the rotation: spring barley, winter barley, winter rape, winter wheat and winter wheat. Typical dates for the field operations are given parentheses.

Activity	S. Barley	W. Barley	W. Rape	W. Wheat	W. Wheat
Plowing	Spring (Mar. 20)	Autumn (Sep. 01)	Autumn (Aug. 15)	Autumn (Sep. 01)	Autumn (Sep. 01)
Seedbed preparation & sowing	Spring (Apr. 05)	Autumn (Sep. 10)	Autumn (Aug. 20)	Autumn (Sep. 10)	Autumn (Sep. 10)
Harvest	Ripe	Ripe	Ripe	Ripe	Ripe
Straw removal	Yes	Yes	No	No	No
Fertilization	121 kg N/ha (Apr. 05)	60 kg N/ha (Mar. 25)	25 kg N/ha (Aug. 20)	55 kg N/ha (Apr. 05)	60 kg N/ha (Apr. 05)
		81 kg N/ha (Apr. 25)	130 kg N/ha (Mar. 05)	74 kg N/ha (May 05)	101 kg N/ha (May 05)

It is recommended to build the management practices for each crop individually as in exercise 1 for the spring barley. Note that winter wheat following a winter rape and winter wheat following a winter wheat are not identical. When a management practice for each of the individual crops has been build then the crop rotation can be build as illustrated in box 6.1.

Simulate the suggested arable farm rotation (only mineral fertilizer) beginning the simulation period January 1st 1994. While running the simulation, try to guess when the simulation will end, i.e. the year the second winter wheat is harvested. Does the simulation run over time? If so, check that you always harvest the last crop before sowing the next.

7 Organic Fertilizer

The objective of this exercise is to introduce the use of organic fertilizer in the simulation of a rotation.

Modify the setup of exercise 6 in order to simulate a pig farm rotation where the main source of fertilizer is pig slurry. The fertilization practice is shown in table 7.1.

The present version of the Daisy model has a deficiency in the way it treats organic fertilizers that are surface broadcasted. The mineral part of the organic fertilizer will eventually enter the soil and can subsequently be utilized by the plants, but the organic part will stay at the surface until the next soil tillage operation, and while staying at the surface no turnover takes place. To compensate for this it is recommended to incorporate the applied organic fertilizer into the upper few centimetres of the soil. Box 7.1 shows how this can be modelled. For further information see *Daisy Program Reference Manual* and *Daisy Tutorial*. Box 7.1 indicates that a pig slurry (name: `pig_slurry`) already has been defined (see “`fertilizer.dai`”). How the volatilization is set can be found in *Daisy Program Reference Manual*. Look up “volatilization” in the index.

Box 6.1: Rotation.

```
;; Spring Barley setup for an arable farm rotation.
(defaction sbarley_A_F_R activity
  (wait_mm_dd 3 20)(plowing)
  (wait_mm_dd 4 05)(fertilize (N25S (weight 121 [kg N/ha])))
  (seed_bed_preparation)(sow "Spring Barley")
  (wait (or (crop_ds_after "Spring Barley" 2.0)(mm_dd 08 30)))
  (harvest "Spring Barley" (stub 8 [cm])(stem 1.0 [])(leaf 1.0 [])))

;; Winter Barley setup for an arable farm rotation.
(defaction wbarley_A_F_R activity
  ;; Insert management actions here.
)

;; Winter Rape setup for an arable farm rotation.
(defaction wrape_A_F_R activity
  ;; Insert management actions here.
)

;; Winter Wheat setup for an arable farm rotation (after a rape).
(defaction wwheat_A_F_R_1 activity
  ;; Insert management actions here.
)

;; Winter Wheat setup for an arable farm rotation (after a cereal).
(defaction wwheat_A_F_R_2 activity
  ;; Insert management actions here.
)

;; Building an arable farm rotation.
(defaction ArableFarmRotation activity
  sbarley_A_F_R wbarley_A_F_R wrape_A_F_R wwheat_A_F_R_1 wwheat_A_F_R_2)
```

Table 7.1: Fertilization practices of the pig farm rotation: spring barley, winter barley, winter rape, winter wheat and winter wheat. Typical dates for the fertilization are given parentheses. Ammonium loss during application is given in squared brackets.

Activity	S. Barley	W. Barley	W. Rape	W. Wheat	W. Wheat
Plowing	Spring (Mar. 05)	Autumn (Sep. 01)	Autumn (Aug. 15)	Autumn (Sep. 01)	Autumn (Sep. 01)
Fertilization (pig slurry)	30 T.w.w./ha (Mar. 05) [5%]	18 T.w.w./ha (Apr. 20) [10%]	20 T.w.w./ha (Aug. 15) [5%] 20 T.w.w./ha (Mar. 20) [10%]	18 T.w.w./ha (Apr. 20) [10%]	18 T.w.w./ha (Apr. 20) [10%]
Fertilization (mineral fertilizer)		68 kg N/ha (Mar. 25)	24 kg N/ha (Mar. 05)	63 kg N/ha (May 05)	95 kg N/ha (Apr. 05)

Box 7.1: Defining an organic fertilizer.

```
;; Defining a pig slurry where 5% of the ammonium content is lost at
;; application (defam = defining added matter)
(defam "Slurry_Vol05%" pig_slurry
  (volatilization 5 [%])
  ;; Other parameters.
)

;; Spring Barley setup for a pig farm rotation.
(defaction sbarley_P_F_R activity
  (wait_mm_dd 3 05)
  (fertilize ("Slurry_Vol05%" (weight 30 [Mg w.w./ha]))(to --5 [cm]))
  (plowing)
  (wait_mm_dd 4 05)
  (seed_bed_preparation)
  (sow "Spring Barley")
  (wait (or (crop_ds_after "Spring Barley" 2.0)(mm_dd 08 30)))
  (harvest "Spring Barley" (stub 8 [cm])(stem 1.0 [ ])(leaf 1.0 [ ])))
```

When inspecting the simulation results please notice the temporal development in soil microbial biomass pools. Does this development look reasonable?

8 Warm-Up Period

The objective of this exercise is to introduce a warm-up period, the purpose of which is to make the simulation less sensitive to the initial conditions.

Repeat the simulation of exercise 7 with the modification that a warm-up period is inserted in front of the simulation (1994–1998). It is a rule of thumb to select one rotation or at least 4 years of simulation as a warm-up period. Hence in the present case the simulation should begin January 1st 1989. Be sure to repeat the rotation; an example can be found in section 5.1.1 of *Daisy Tutorial*.

Did the warm-up period improve the credibility of the simulation?

9 The MIT Model

The objective of this exercise is to calibrate the MIT-model (Mineralization-Immobilization Turnover).

In box 1.3 organic matter section was introduced. The total carbon content of the soil is calculated from the humus content, which was defined in the horizon section (box 1.2). The MIT model comprises several organic matter pools and the total carbon content is distributed among these pools by the initialization procedure that makes use of annual input of organic carbon. This, in combination with a warm-up period, will often give a reasonable initialization and hence a reasonable N-mineralization. However this can be fine tuned by introducing the `background_mineralization` init parameter, like this:

```
(OrganicMatter original
  (init (input 3000 [kg C/ha/y])
    (background_mineralization 30 [kg N/ha/y])))
```

Daisy will then attempt to initialize the organic matter pools so the total mineralization is 30 kg N/ha/y higher than the amount of organic bound nitrogen added to the system together with the 3000 kg C/ha/y we specified above. In other words, at the time we start the simulation the amount of organic bound nitrogen in the soil is expected to decrease (be released) at a rate of 30 kg N/ha/y.

In certain cases the model may not be able to initialize itself, and it may be necessary to park some of the soil organic carbon in an inactive pool. This is done by entering the following parameter in the `defhorizon` section

```
(SOM_fractions -1 -1 xxx)
```

where *xxx* is the inactive fraction of the soil organic carbon.

A way to estimate the N mineralization level of the system is to grow an unfertilized plot and measure the nitrogen content of the crop at harvest. Consider the agro-ecosystem of exercise 8 and assume that in 1993 and 1994 an unfertilized plot were grown with winter wheat. The above ground part of the crop was harvested in the two years and the biomass was analyzed for its N content. The harvested nitrogen was 102 kg N/ha and 60 kg N/ha in 1993 and 1994, respectively. Use this information to calibrate the MIT model by modifying the background mineralization parameter, and if necessary, the `SOM_fractions` parameter.

10 Building a Weather File

The objective of this exercise is to build a weather file.

The Daisy weather-file is written in the Daisy weather format (extension **dwf**). In front the weather data is a header as shown in Box 10.1 The first line in the header tells Daisy that the present file is written

Box 10.1: Beginning of weather file.

```
dwf-0.0
Station: Taastrup
Elevation: 30 m
Longitude: 12 dgEast
Latitude: 55 dgNorth
TimeZone: 15 dgEast
Surface: reference
ScreenHeight: 2 m
Begin: 1962-04-01
End: 2001-10-07
NH4WetDep: 0.9 ppm
NH4DryDep: 2.2 kgN/ha/year
NO3WetDep: 0.6 ppm
NO3DryDep: 1.1 kgN/ha/year
TAverage: 7.8 dgC
TAmplitude: 8.5 dgC
MaxTDay: 209 yday
Timestep: 24 hours
PrecipCorrect: 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0
-----
Year Month Day  GlobRad AirTemp Precip
year month mday W/m^2  dgC      mm/d
1962 4      1    120.4   2.8      0.0
1962 4      2    130.7   4.4      5.3
1962 4      3    39.1    6.8      2.9
... more climate data here ...
```

in the Daisy weather format version 0.0 (the only version implemented so far). Then follows a number of keywords: **Station**, **Elevation**, **Longitude**, **Latitude**, and **TimeZone**; defining name and location of the weather station where the data were measured. **Surface**: reference refers to the surface above which the weather data was measured (reference is short grass common for most weather stations). **ScreenHeight** gives the height at which air temperature (and humidity and wind speed if measured) was recorded. **Begin** and **End** defines the date of the first and last record in the dataset. **NH4WetDep**, **NH4DryDep**, **NO3WetDep**, and **NO3DryDep** define deposition parameters at the location. **TAverage**, **TAmplitude**, and **MaxTDay** define the thermal regime of locality (annual average air temperature, the amplitude of the annual air temperature cycle, and the day of the year when maximum air temperature is expected on average). The **Timestep** keyword tells the model basic time-step of the data, in the present case we are dealing with daily values. **PrecipCorrect** gives twelve correction factors used to correct the precipitation month by month. In the present case the precipitation is measured at the soil surface,

hence no correction is needed for the aerodynamic effect. The next three lines form the header for the actual weather data, which in the present example are written in a TAB separated format.

The present example shows a minimum weather data required. However, Daisy can make use of more detailed data (see *Daisy Program Reference Manual*).

Build a Daisy weather file based on the data in the spreadsheet **AArslevWeatherDat.xls**. Årslev is a locality located at the center of Funen. The precipitation is measured at 1.5 m height and the shelter category is B. The surface of weather station is short grass and the screen height is 2 m. The precipitation correction is found in table 10.1

Table 10.1: Correction of point precipitation as function of shelter categories [per cent], [Allerup et al., 1998].

Shelter	J	F	M	A	M	J	J	A	S	O	N	D	Year
A	29	30	26	19	11	9	8	8	9	10	17	26	16
B	41	42	35	24	13	11	10	10	11	14	23	37	21
C	53	53	45	29	16	13	12	12	13	17	29	48	27

11 A New Organic Fertilizer Parameterization

The objective of this exercise is to introduce a new organic fertilizer in fertilizer library.

The composition of a mixed slurry is: dry matter content 4.8%; carbon content 36% (dry matter basis); nitrogen content 10.5 % (dry matter basis); ammonium content 8.5% (dry matter basis); and nitrate content 0.0% (dry matter basis). A batch experiment has shown that the mineralization of the mixed slurry approximately can be described by splitting the organic matter into two pools and assume first order degradation kinetics for the pools. The resistant pool constituted 79% of the C with a half-life of 150 days and the easier decomposable pool constituted 21% of the C with a half-life of 15 days. From previous experience it was suggested that the C/N of the easy decomposable pool should be around 5.

Include the mixed slurry in the fertilizer library. Help on how to write a new parameterization of a new organic fertilizer can be found in the *Daisy Program Reference Manual*, section 17.3 and 8.

Compare the results obtained with the new mixed slurry with the results obtained in exercise 8.

12 Crop Parameterization

The objective of this exercise is to parameterize a maize crop.

The maize parameters found in the crop-file “**PioneerMaize.dai**” are mainly based on [de Vries et al., 1989]. However, the measured data of LAI, and leaf, stem, and cob weight from 1998 found in the spreadsheet “**MaizeProd.xls**” are from a silage maize. Define a new maize by assuming that the silage maize parameters partly resembles the Pioneer Maize parameters. Calibrate selected parameters making use of the measured data. When a new slightly different parameterization is defined it is recommended not copying parameter settings and making changes, but instead to derive the new parameterization from the existing one. This makes it easy to see where the parameterizations differ.

The main elements of a crop definition are shown in box 12.1. The crop definition comprises a number of subsections: **Devel** contains parameters pertaining to the phenological development of the crop; **LeafPhot** parameters govern the leaf photosynthesis; **Canopy** parameters are describing LAI development and distribution; **Root** contains root penetration and root distribution parameters; **Partit** governs the partition of assimilates; **Prod** are dealing with respiration and production parameters; **CrpN** parameters

Box 12.1: Crop definition.

```
(defcrop "Pioneer Maize (WS)" default
;; Phenological development :
;; Emergence    DS = 0
;; Flowering    DS = 1
;; Maturity     DS = 2
(Devel default
;; Temperature sum calculated from sowing to emergence.
(EmrTSum 350)
;; Development rate [d-1] in the vegetative period
(DSRate1 0.0265)
;; Effect of temperature on development rate in the veg. period
(TempEff1 (8. 0.00) (10. 0.30) (15. 0.75) (25. 1.00) (35. 1.20))
;; Effect of photoperiod on development rate in the veg. period
(PhotEff1 (0. 0.90) (12. 1.00) (14. 0.95) (16. 0.90) (24. 0.90))
;; Development rate [d-1] in the reproductive period
(DSRate2 0.017)
;; Effect of temperature on development rate in the rep. period
(TempEff2 (8. 0.00) (10. 0.30) (15. 0.75) (25. 1.00) (35. 1.20)))
(LeafPhot
;; Insert parameters here.
)
(Canopy
;; Insert parameters here.
)
(Root
;; Insert parameters here.
)
(Partit
;; Insert parameters here.
)
(Prod
;; Insert parameters here.
)
(CrpN
;; Insert parameters here.
)
(Harvest
;; Insert parameters here.
))
```


are controlling the plant relations; and **Harvest** parameters are dealing with the harvest itself as well as the fate of plant residuals. All crop parameters are defined in *Daisy Program Reference Manual*. When consulting the manual it is noticed that more than one crop model is implemented in the Daisy code. However, in this case we use the standard crop model, which is the default crop model. This crop model resembles the SUCROS model [van Keulen et al., 1982].

Parameters can be given in two ways: 1. as a single value (e.g. **DSRate1**); 2. as a piecewise linear function (e.g. **TempEff1**). The piecewise linear function is defined by a number of points (x,y) on the considered curve. In the example, **TempEff1**, x equals temperature and y equals temperature effect. In the crop definition it is common to make a parameter a function of the development stage (DS). The only exception is the subsection **Devel**, where the parameters governing the DS are defined. Notice that **Devel** requires the keyword **default** because more than one model describing the development is implemented.

We have no information on roots, plant nitrogen dynamics or plant residues. Hence, in the calibration we need not consider the **Root**, **CrpN** (the nitrogen supply is ample so we need not consider N-stress), and **Harvest** sections. We have no reason to believe that the present maize variety should differ from the pioneer maize in respect to respiration, hence we may want to keep these parameters without change (keeping the **Prod**-parameters). The relevant **Canopy**-parameters in this context are pertaining to the leaf area. The specific leaf area (LAI per unit leaf weight) can be calculated directly from measured data (it is noticed that the value is not constant during the growth period, however the model can take this into account, see *Daisy Program Reference Manual*).

If we want to change the development rate of the maize in the vegetative period this can be done by writing:

```
(defcrop "Exercise 12 Maize" "Pioneer Maize (WS)"
  (Devel original
    (DSRate1 0.0160)))
```

The keyword **original** retains the original parameterization, hence only the **DSRate1** is changed.

In this way the selected parameters can be changed one by one in order to fit the data. We may have to do quite a number of simulations when calibrating the model. In order to save time it is possible to make use of the checkpoint facility (see *Daisy Tutorial*). By using this facility it is possible to write a checkpoint at a predefined time and then later continue the simulation from that point in time. In the present example we run the warm-up period and then make a checkpoint just before the sowing of the maize (e.g. (**checkpoint (when (at yyyy mm dd hh))**)). The checkpoint is to be inserted among the log files in the output section. Another facility which also often is helpful is the (**activate_output (after yyyy mm dd hh)**), which is used to suppress output during warm-up periods.

Use the calibrated setup of exercise 9 and assume that the maize is sown the year after the second year winter wheat. 50 ton w.w./ha was surface broadcasted and plowed in 27. April 1998. Maize was sown 4. May and 142 kg N/ha in ammonium nitrate was applied the same day. The Maize was harvested for silage 3. November, however it was not ripe at that date.

In the calibration procedure it is advisable to begin with the DS. We have no information of when the different growth stages of the maize occurred. But we have information on the onset of growth of the cobs and we can use this in order to calibrate the development rate in the vegetative period. We have no information that makes it possible to do a similar calibration for the reproductive period.

It is recommended that when the DS is calibrated we should continue looking at the **Canopy**-parameters and afterwards at the **LeafPhot**-parameters and **Partit**-parameters.

A Standard log files

The columns for the main log files.

A.1 harvest

Create a log of all harvests.

Table columns (common):

- *stem_DM*: [Mg DM/ha]
Stem dry matter removed by harvest.
- *dead_DM*: [Mg DM/ha]
Yeallow leaves dry matter removed by harvest.
- *leaf_DM*: [Mg DM/ha]
Green leaves dry matter removed by harvest.
- *sorg_DM*: [Mg DM/ha]
Storage organ (grains or tuber) dry matter removed by harvest. For some crops, only the economically important part of the storage organ is counted.

Table columns (if print_N is set):

- *stem_N*: [kg N/ha]
Stem nitrogen removed by harvest.
- *dead_N*: [kg N/ha]
Yeallow leaves nitrogen removed by harvest.
- *leaf_N*: [kg N/ha]
Green leaves nitrogen removed by harvest.
- *sorg_N*: [kg N/ha]
Storage organ (grains or tuber) nitrogen removed by harvest.

Table columns (if print_C is set):

- *stem_C*: [kg C/ha]
Stem carbon removed by harvest.
- *dead_C*: [kg C/ha]
Yeallow leaves carbon removed by harvest.
- *leaf_C*: [kg C/ha]
Green leaves carbon removed by harvest.
- *sorg_C*: [kg C/ha]
Storage organ (grains or tuber) carbon removed by harvest.

Table columns (common):

- *WStress*: [d]
Number of days worth of production lost due to water stress.
This is just a rough measure. It is calculated by weighting the water stress variation over the day with the amount of global radiation received in the same period. So water stress at noon with high radiation counts a lot more than water stress near sunset with low radiation. All days are counted the same, so water stress a day with low radiation is counted the same as water stress a day with high radiation.
- *NStress*: [d]
Number of days worth of production lost due to nitrogen stress.
This is just a rough measure. It is calculated by weighting the nitrogen stress variation over the day with the amount of global radiation received in the same period. So nitrogen stress at noon with high radiation counts a lot more than nitrogen stress near sunset with low radiation. All days are counted the same, so nitrogen stress a day with low radiation is counted the same as nitrogen stress a day with high radiation.
- *WP_ET*: [kg/m³]
A measure for water usage efficiency. Specifically the economic yield (storage organ) divided by the total evapotranspiration in the period from emergence to maturity or harvest.

A.2 Crop

A ‘crop’ model defined in ‘log-std.dai’.

Crop development and production.

Table columns:

- *DS*: [DS]
Development stage of the crop. 0 is emergence, 1 is flowering, 2 is ripe.
- *Height*: [cm]
Height of crop.
- *LAI*: [m²/m²]
Leaf area index.
- *Depth*: [cm]
Effective depths of root zone.
- *WLeaf*: [Mg DM/ha]
Dry matter content of leaves.
- *WDead*: [Mg DM/ha]
Dry matter content of dead (yellow) leaves still on crop.
- *WStem*: [Mg DM/ha]
Dry matter content of stem.
- *WSOrg*: [Mg DM/ha]
Dry matter content of storage organ (grain, tuber).

- *WRoot*: [**Mg DM/ha**]
Dry matter content of roots.
- *Fixated*: [**kg N/ha**]
Atmospheric nitrogen fixated by the crop (root biome).
- *NLeaf*: [**kg N/ha**]
Nitrogen content of the leaves.
- *NDead*: [**kg N/ha**]
Nitrogen content of dead (yellow) leaves still on the crop.
- *NStem*: [**kg N/ha**]
Nitrogen content of the stem.
- *NSOrg*: [**kg N/ha**]
Nitrogen content of the storage organ (grain or tuber).
- *NRoot*: [**kg N/ha**]
Nitrogen content of the roots.
- *water_stress*: [**<none>**]
Highest water stress experienced in the period.

When the root system cannot fulfill the demand of water from the canopy, the crop experience water stress, and production suffers.

The value is between 0 (no water stress) and 1 (no production).
- *nitrogen_stress*: [**<none>**]
Highest nitrogen stress experienced in the period.

When the crop nitrogen content is below a critical level, crop experience nitrogen stress, and production suffers.

The value is between 0 (no nitrogen stress) and 1 (no production).
- *Phenology*: [**<unknown>**]
Current phenological stage.

For most crops, this is -1 (not yet emerged), 0 (vegetative), 1 (reproductive), or 2 (ripe). Some crops may show other scaled, like BBCH.

A.3 Field water

A 'column' model defined in 'log-std.dai'.

Information about water input, output and content for the system.

The intended use of this log is large scale water balance, for example reservoir management. It provide information about how much water is in the field (down to a specified depth), where it is located in the field (surface, soil matrix or in biopores), as well as the sources, sinks and amounts of water entering or leaving the system. It does not provide information about internal translocation of water between surface, soil matrix and biopores, use see the 'Soil water' log instead for that.

For the balances of this log to work, you must include the entire root zone, as well as the biopore zone.

Table columns:

- *Precipitation*: [mm]
Total amount of water entering the system as rain and snow.
- *Irrigation*: [mm]
Total amount of water added to the system from irrigation.
- *Potential evapotranspiration*: [mm]
There is energy enough to evaporate this amount of water.
- *Actual evapotranspiration*: [mm]
Amount of water removed by evaporation and transpiration.
The evaporation part covers water evaporated from the soil surface, the snow pack, or intercepted water on the canopy. The transpiration covers water removed from the soil through the root system and the stomata on the leaves.
The actual evapotranspiration will be lower than the potential evapotranspiration when the available water is insufficient.
- *Matrix percolation*: [mm]
This is the amount of water leaving the system through the soil bottom. It can be negative if there is capillary rise.
- *Biopore percolation*: [mm]
This is the amount of water leaving the system through biopores.
- *Matrix drain flow*: [mm]
Amount of water leaving the system through the soil matrix to drain pipes.
Note that the total amount of water in the drain pipes may be higher than this if you have specified the 'to' parameter, as the whole soil profile will contribute to the drain flow, and only the contributions from the soil interval being logged is counted here.
- *Biopore drain flow*: [mm]
Amount of water leaving the system through biopores to drain pipes.
Note that the total amount of water in the drain pipes may be higher than this if you have specified the 'to' parameter, as the whole soil profile will contribute to the drain flow, and only the contributions from the soil interval being logged is counted here.
- *Runoff*: [mm]
Amount of water running off the surface.
This is intended to simulate water runoff from a surface with a slope. However, since Daisy is a one dimensional model, the water has nowhere to go. For typical setups, this value will be zero.
- *Biopore water*: [mm]
The total amount of biopore water in the soil interval
- *Soil matrix water*: [mm]
Total water content of the soil.
- *Surface water*: [mm]
Total water stored on the surface. This includes the snow pack, intercepted water, ponding, and water in the litter layer.

A.4 Soil water

A 'column' model defined in 'log-std.dai'.

Information about water input, output and content for the soil.

The intended use of this log is small scale water management, for example irrigation optimization. Above ground water is considered external to the system. The balances provided by the log will work on any subset of the soil that matches the discretization.

Table columns:

- *Matrix infiltration*: [mm]
Amount of water entering the soil through the matrix system.
This is the normal infiltration channel.
- *Matrix percolation*: [mm]
Amount of water leaving the soil through the matrix system.
This is the normal percolation channel.
- *Biopore infiltration*: [mm]
Amount of water entering the soil through the biopores.
- *Biopore percolation*: [mm]
Amount of water leaving the soil through the biopore system.
- *Subsoil irrigation*: [mm]
Amount of water incorporated directly into the soil.
This includes both subsoil irrigation and any water in fertilizer directly incorporated into the soil.
- *Matrix direct drain flow*: [mm]
Flow from matrix directly to the drain pipes.
This does not count water that flows to the drain pipes through biopores.
Note that water flows to the drain pipes from all soil layers, both above and below the pipes, and even from the automatically inserted aquitard horizon that lies below the normal soil horizons. So if you have specified less than the full soil profile, that is, if you have set the 'from' or 'to' parameters, the amount you see logged here will be less than the total amount of water in the drain pipes.
If the soil has no drain pipes this amount will be zero.
- *Matrix indirect drain flow*: [mm]
Flow from matrix indirectly to the drain pipes.
This only counts water that flows to the drain pipes through biopores.
- *Biopore drain flow*: [mm]
Water in the interval that flows from the biopores to the drain pipes.
Note that water flows to the drain pipes from all soil layers, both above and below the pipes, and even from the automatically inserted aquitard horizon that lies below the normal soil horizons. So if you have specified less than the full soil profile, that is, if you have set the 'from' or 'to' parameters, the amount you see logged here will be less than the total amount of water in the drain pipes.

If the soil has no biopores connected directly to the drain, this amount will be zero.

- *Root extraction:* [mm]

Water in the soil interval extracted by the roots.

This number will be equal to the transpiration if the soil interval being logged includes the entire root zone.

- *Freezing:* [mm]

Water in the soil interval turned into ice.

This number will be zero unless the 'enable_ice' SoilHeat parameter has been set.

- *Tillage:* [mm]

Net amount of water being added to the soil interval by tillage operations.

This number will be zero if the soil interval includes the all the soil affected by the tillage operation, which is the normal case. The number will be negative if more water have been added than removed by the tillage operation.

- *Matrix to biopores:* [mm]

Amount of water leaving the soil matrix to the biopores. This does not include water that flows to drain connected biopores.

- *Biopores to matrix:* [mm]

Amount of water leaving the biopores to the matrix.

- *Matrix water:* [mm]

The total amount of soil matrix water in the soil interval

- *Biopore water:* [mm]

The total amount of biopore water in the soil interval

A.5 Field nitrogen

A 'column' model defined in 'log-std.dai'.

Nitrogen input, output, transformation and content for the system.

The intended use of this log is large scale nitrogen balance, for example reservoir management. It provide information about how much nitrogen is in the field (down to a specified depth), where it is located in the field (surface, soil matrix or in biopores), what form is has (crop, soil organic matter, or mineral) as well as the sources, sinks and amounts of nitrogen entering or leaving the system, and transformation between the four forms. It does not provide information about internal translocation of nitrogen between surface, soil matrix and biopores, use see the 'Soil nitrogen' log instead for that.

For the balances of this log to work, you must include the entire root zone

Table columns:

- *Min-Surface-Fertilizer:* [kg N/ha]

Mineral fertilizer added above the soil surface.

- *Min-Soil-Fertilizer:* [kg N/ha]

Mineral fertilizer incorporated into the soil.

- *Deposition*: [kg N/ha]
Nitrogen added to the soil surface as atmospheric deposition.
- *Matrix-Leaching*: [kg N/ha]
Loss from leaching through the soil matrix below the specified depth. This can be negative in case of capillary rise.
- *Biopore-Leaching*: [kg N/ha]
Loss from leaching through biopores below the specified depth.
- *Soil-Drain*: [kg N/ha]
Loss of nitrogen from the soil matrix to drain pipes.
- *Biopore-Drain*: [kg N/ha]
Loss of nitrogen from the biopores to drain pipes.
- *Surface-Loss*: [kg N/ha]
Loss of mineral nitrogen stored above surface.

This includes volatilization, surface runoff, mineral nitrogen on leaves removed at harvest, and chemical transformations.
- *Min-Surface*: [kg N/ha]
Mineral nitrogen stored above the soil surface. This includes leaves, ponded water, etc.
- *Min-Soil*: [kg N/ha]
Mineral nitrogen stored in the soil matrix.
- *Min-Biopores*: [kg N/ha]
Mineral nitrogen stored in the biopores.
- *Error*: [kg N/ha]
Mineral nitrogen 'borrowed from the future'.

This can be non-zero if the processes that remove nitrogen from some place in the soil are fast compared to the timestep. If this happens, you should rerun the simulation with a shorter timestep. Or readjust the system to slow down those processes that remove mineral nitrogen.
- *Mineralization*: [kg N/ha]
NH₄ produced in the soil matrix by mineralization of organic matter.
- *Immobilization*: [kg N/ha]
NO₃ immobilized as a result of organic matter turnover.
- *Crop-Uptake*: [kg N/ha]
Nitrogen uptake by the roots.
- *Volatilization*: [kg N/ha]
NH₄-fertilizer lost during the application.
- *N₂O-Nitrification*: [kg N/ha]
N₂O production in the soil matrix due to nitrification.

The N₂O produced is assumed to escape to the atmosphere, and will not be traced by the model.

- *NO3-Denitrification*: [kg N/ha]
Loss of NO3-N due to denitrification.
- *N2O-Denitrification*: [kg N/ha]
N2O-N released by denitrification.
- *Fixated*: [kg N/ha]
Atmospheric nitrogen fixated by the crop.
- *Org-Fertilizer*: [kg N/ha]
Organically bound nitrogen supplied by fertilizers.
- *Seed*: [kg N/ha]
Nitrogen supplied in seeds when sowing.
- *Harvest*: [kg N/ha]
Nitrogen removed as part of the harvest.
- *Residuals-Surface*: [kg N/ha]
Above ground plant residuals.
This includes both exfoliation during crop growth, and any above ground residuals left after harvest.
- *Residuals-Soil*: [kg N/ha]
Below ground plant residuals.
This includes root death and rhizodeposition during plant growth, as well as the roots left after harvest.
- *Org-Surface*: [kg N/ha]
Nitrogen content of organic material on the soil surface.
This does not include nitrogen in the crop.
- *Org-Soil*: [kg N/ha]
Nitrogen content of organic material below the soil surface.
This does not include nitrogen in living roots.
- *Crop*: [kg N/ha]
Nitrogen content of the crop, excluding dead leaves.

A.6 Soil nitrogen

A ‘column’ model defined in ‘log-std.dai’.

Nitrogen input, output, transformation and content for the soil.

Table columns:

- *NO3-Matrix-In*: [kg N/ha]
NO3-N infiltration.
- *NO3-Biopore-In*: [kg N/ha]
NO3-N infiltration.

- *NO3-Leak-Matrix*: [kg N/ha]
NO3-N leaching.
- *NO3-Leak-Biopore*: [kg N/ha]
NO3-N leaching.
- *NO3-to-Biopores*: [kg N/ha]
Movement of NO3-N from soil matrix to biopores.
- *NO3-from-Biopores*: [kg N/ha]
Movement of NO3-N from soil matrix to biopores.
- *NO3-Soil-Drain*: [kg N/ha]
NO3-N lost to drains directly from the soil matrix.
- *NO3-Indirect-Drain*: [kg N/ha]
NO3-N lost to drains from the soil matrix, through biopores.
- *NO3-Biopore-Drain*: [kg N/ha]
NO3-N lost to drains from biopores.
- *NO3-Incorp*: [kg N/ha]
NO3-N incorporated directly into the soil.
- *NO3-Tillage*: [kg N/ha]
NO3-N added by tillage operations.
- *NO3-Uptake*: [kg N/ha]
NO3-N removed by plant roots.
- *NO3-Soil*: [kg N/ha]
NO3-N content of the soil matrix (excluding biopores).
- *NO3-Biopores*: [kg N/ha]
NO3-N content of the biopores.
- *NO3-Error*: [kg N/ha]
NO3-N borrowed from the future.

This may be temporarily non-zero if some process remove NO3 too fast compared to the time step.
- *NH4-Matrix-In*: [kg N/ha]
NH4-N infiltration.
- *NH4-Biopore-In*: [kg N/ha]
NH4-N infiltration.
- *NH4-Leak-Matrix*: [kg N/ha]
NH4-N leaching.
- *NH4-Leak-Biopore*: [kg N/ha]
NH4-N leaching.

- *NH₄-to-Biopores*: [kg N/ha]
Movement of NH₄-N from soil matrix to biopores.
- *NH₄-from-Biopores*: [kg N/ha]
Movement of NH₄-N from soil matrix to biopores.
- *NH₄-Soil-Drain*: [kg N/ha]
NH₄-N lost to drains directly from the soil matrix.
- *NH₄-Indirect-Drain*: [kg N/ha]
NH₄-N lost to drains from the soil matrix, through biopores.
- *NH₄-Biopore-Drain*: [kg N/ha]
NH₄-N lost to drains from biopores.
- *NH₄-Incorp*: [kg N/ha]
NH₄-N incorporated directly into the soil.
- *NH₄-Tillage*: [kg N/ha]
NH₄-N added by tillage operations.
- *NH₄-Uptake*: [kg N/ha]
NH₄-N removed by plant roots.
- *NH₄-Soil*: [kg N/ha]
NH₄-N content of the soil matrix (excluding biopores).
- *NH₄-Biopores*: [kg N/ha]
NH₄-N content of the biopores.
- *NH₄-Error*: [kg N/ha]
NH₄-N borrowed from the future.

This may be temporarily non-zero if some process remove NH₄ too fast compared to the time step.
- *NO₃-Denitrification*: [kg N/ha]
NO₃-N removed by denitrification.
- *N₂O-Denitrification*: [kg N/ha]
N₂O-N released by denitrification.
- *NH₄-Nitrification*: [kg N/ha]
NH₄-N removed by nitrification.
- *NO₃-Nitrification*: [kg N/ha]
NO₃-N added by nitrification.
- *N₂O-Nitrification*: [kg N/ha]
NH₄-N lost as N₂O as a byproduct of the nitrification process.

NH₄-Nitrification = NO₃-Nitrification + N₂O-Nitrification

- *NH₄-Mineralization*: [kg N/ha]
NH₄-N added from mineralization of organic matter in the soil.
- *NO₃-Immobilization*: [kg N/ha]
NO₃-N removed by the organic matter turnover processes.
- *Residuals-N*: [kg N/ha]
Nitrogen in organic matter added to the soil from roots.
- *Tillage-Org_N*: [kg N/ha]
Nitrogen in organic matter added to the soil by tillage operations.
- *Bioincorporation*: [kg N/ha]
Nitrogen in organic matter added to the soil by bioincorporation.
This includes dead leaves removed from the soil surface by earthworms.
- *AOM*: [kg N/ha]
Total amount of nitrogen in added organic matter in the soil.
AOM consists of fertilizer and plant residuals that has not yet been decomposed.
- *SOM*: [kg N/ha]
Total amount of nitrogen in soil humus.
This consist of dead organic matter that can no longer be traced to its origins.
- *SMB*: [kg N/ha]
Nitrogen in soil microbiological organisms.
The living part of the soil (excluding plant roots).
- *Buffer*: [kg N/ha]
Nitrogen in humus added by fertilizer but not yet available for turnover.
This is usually zero.

A.7 Field chemical

A ‘chemical’ model defined in ‘log-std.dai’.

Content, transport and transformation of chemicals in the field.

Table columns:

- *Spray*: [g/ha]
Applied to the surface.

Chemicals can be added to the surface through the spray, fertilize and irrigate management operations.
- *Deposit*: [g/ha]
Athmospheric deposition.

This is usually zero when tracking pesticides, but can be relevant for atmospheric polutants, and is also significant when for nitrogen.

- *Harvest*: [g/ha]
Removed by harvest.
Some amount of the chemical may be left on the canopy, and removed during harvest. It is usually not significant for the overall balance, but might be important to the consumers, depending on the chemical.
- *Dissipate*: [g/ha]
Dissipating from the canopy.
Some chemicals evaporate or decompose when stored on the canopy.
- *Litter Decompose*: [g/ha]
Decomposed while stored in the litter pack.
- *Litter Transform*: [g/ha]
Added from chemical transformation in the litter pack.
- *Surface Decompose*: [g/ha]
Decomposed while stored on the soil surface.
- *Surface Transform*: [g/ha]
Added from chemical transformation on the soil surface.
- *Runoff*: [g/ha]
Lost from soil surface due to lateral water flow.
- *Leak matrix*: [g/ha]
Lost due to leaching.
- *Leak biopores*: [g/ha]
Lost due to leaching.
- *Biopore drain*: [g/ha]
Lost from biopores to drain pipes.
- *Soil drain*: [g/ha]
Lost from the soil matrix to drain pipes.
- *Surface drain*: [g/ha]
Amount moving directly from surface to drain, through biopores.
This never come in contact with the soil matrix.
- *External*: [g/ha]
Added to the soil through some external mechanism, such as subsoil irrigation.
- *Uptake*: [g/ha]
Removed from the soil matrix through plant roots.
- *Soil Decompose*: [g/ha]
Decomposed while in the soil matrix.
- *Soil Transform*: [g/ha]
Added through chemical transformation in the soil matrix.

- *Snow*: [g/ha]
Amount stored in the snow pack.
- *Canopy*: [g/ha]
Amount stored in on the canopy.
- *Litter*: [g/ha]
Amount stored in the surface litter (mulch, residuals, fertilizer).
- *Surface*: [g/ha]
Amount stored on the soil surface.
- *Soil*: [g/ha]
Amount stored in the soil matrix.
- *Biopores*: [g/ha]
Amount stored in the biopores.
- *Error*: [g/ha]
Amount borrowed from the future.

This can be temporarily non-zero if some process remove the chemical fast compared to the timestep of the simulation.

A.8 Soil chemical

A ‘chemical’ model defined in ‘log-std.dai’.

Content, transport and transformation of chemicals in the soil.

Table columns:

- *In-Matrix*: [g/ha]
Infiltration through soil matrix.
- *In-Biopores*: [g/ha]
Infiltration through biopores.
- *Leak-Matrix*: [g/ha]
Leaching through the soil matrix.
- *Leak-Biopores*: [g/ha]
Leaching.
- *Biopores to matrix*: [g/ha]
Enter the soil matrix from the biopores.
- *Matrix to biopores*: [g/ha]
Enter the biopores from the matrix.
- *Tillage*: [g/ha]
Added through tillage operations.

- *Drain-Soil*: [g/ha]
Lost to drains directly from the soil matrix. This does not include what enters the drains through the biopores.
- *Drain-Biopores*: [g/ha]
Lost to drains from the biopores.
- *Drain-Biopores-Indirect*: [g/ha]
Lost to drains from matrix through biopores. This does not include what enters the drain directly from the soil matrix.
- *External*: [g/ha]
Added externally, for example through subsoil irrigation.
- *Uptake*: [g/ha]
Lost with water uptake by plant roots.
- *Decompose*: [g/ha]
Decomposed.
- *Transform*: [g/ha]
Added by chemical transformation.
- *Content*: [g/ha]
Total amount in soil matrix in the specified soil interval.
- *Biopores*: [g/ha]
Total amount in biopores in the specified soil interval.
- *Error*: [g/ha]
Borrowed from the future.

This can be non-zero if some process remove the chemical too fast compared to the simulation timestep.

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